A Feynman–Wigner-Style Diagnostic for the Efficacy of Conformal Prediction via Signed de Finetti Representations

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Draft — companion note

Abstract

Conformal prediction builds prediction sets that cover the truth at a rate you choose, finite-sample and distribution-free, assuming only exchangeable data. That guarantee is marginal. de Finetti's theorem describes the exchangeability it rests on, and in the finite form the mixing measure may be signed (Kerns and Székely, 2006). We present a short lemma that decomposes the slope of conformal's calibration-conditional coverage into two terms. One is a non-negative threshold term, the classical Beta-law fan. The other carries the sign of the de Finetti measure. Positive (extendable) mixtures make conformal conditionally adaptive. The signed corner—demeaned, ranked, or compositional scores—makes it anti-adaptive. The marginal guarantee is the same either way. Only what it hides changes.

1 Setup

Split conformal returns a set $C(X)$ with $\mathbb{P}\big(Y \in C(X)\big) \geq 1-\alpha$ for any distribution and any sample size, provided the data are exchangeable. That marginal guarantee is the whole appeal, and it is not in dispute here. The question is what it says about the case in front of you.

Scores $S_1, \ldots, S_n$ (calibration) and S_{n+1} (test) are jointly exchangeable with a continuous joint law (no ties). The conformal width is the order statistic $\hat{q} = S_{(k)}$ of the calibration scores, $k = \lceil (n+1)(1-\alpha) \rceil$, and a test point is covered when $S_{n+1} \leq \hat{q}$.

Two kinds of dependence live in any such problem, and they should not be confused. *Within a row*: inside a single (x, y), does the residual's shape depend on x? This is the subject of the companion paper (Cotton), where the log-score regret of a single-shape conformal system equals the mutual information $I(R; X)$. It is a property of one observation's law and is present even when the rows are i.i.d. *Across rows*: how do distinct observations relate to one another—independent, sharing a latent cause, or mutually constrained? This is the subject of de Finetti's theorem, and of this note. The two axes are orthogonal. This note is entirely about the second.

2 Finite de Finetti and the sign of the mixture

de Finetti's theorem pins down what infinite exchangeability is.

Theorem 1 (de Finetti; de Finetti, 1937; Hewitt and Savage, 1955)**.** *Let $X_1, X_2, \ldots$ be an infinitely exchangeable sequence in a Polish space. Then there is a probability measure π on the space of distributions such that, for every n, the law of $(X_1, \ldots, X_n)$ equals $\int P^{\otimes n}\, \mathrm{d}\pi(P)$. Equivalently, conditional on a random measure $\Theta \sim \pi$, the X_i are i.i.d. with law Θ.*

The mixing measure π is a genuine prior, and that forces a sign. A positive mixture has $\mathrm{Cov}(S_i, S_j) = \mathrm{Var}\left(\mathbb{E}[S_i \mid \Theta]\right) \geq 0$, so de Finetti mixtures are non-negatively correlated. Finite exchangeable sequences are not so constrained. Exchangeability alone forces only $\rho \geq -1/(n-1)$ (Aldous, 1985), and the negative band is real—sampling without replacement sits at its floor.

The finite theory completes the picture. Diaconis and Freedman (1980) show a finite exchangeable sequence is *approximately* a positive i.i.d. mixture, within total variation $O(k/n)$. Kerns and Székely (2006) make it exact, at the price of signing.

Theorem 2 (Kerns–Székely; Kerns and Székely, 2006)**.** *Let $(X_1, \ldots, X_n)$ be exchangeable on a measurable space. Then there is a finite* signed *measure μ of total mass 1 on the space of distributions with $\mathrm{law}(X_1, \ldots, X_n) = \int P^{\otimes n}\, \mathrm{d}\mu(P)$. One may take $\mu \geq 0$ (a genuine prior, recovering Theorem 1) if and only if the sequence extends to an infinitely exchangeable one (Leonetti, 2018).*

We picture this as the i.i.d. laws tracing a curve: positive mixtures are its convex hull, the extendable $\rho \geq 0$ laws, and signed mixtures its affine span, everything. A signed representation marks non-extendability—a finite exchangeable sequence that cannot be continued to an infinite one—which is the negative-correlation regime.

We frame the signed measure as a *negative probability*, the kind Feynman (1987) used partway through a calculation, or Wigner (1932)'s phase-space distribution, which dips below zero while every observable marginal stays a genuine probability (the idea reached statistics through Bartlett, 1945). You cannot sample a parameter from it. That is why the signed corner gives a diagnosis, not a model: recovering a latent means fitting a real one, not reading the quasi-probability off.

So the sign of the de Finetti mixing measure is a clean label: **positive** = extendable, non-negatively correlated, a genuine prior. **Signed** = finite, negatively correlated, no genuine prior.

3 Marginal coverage ignores the sign

For continuous exchangeable scores the rank of S_{n+1} among all $n+1$ is uniform, and $S_{n+1} \leq S_{(k)}$ iff that rank is at most k, so

$$\mathbb{P}\big(S_{n+1} \leq \hat{q}\big) = \frac{k}{n+1} \geq 1 - \alpha. \tag{1}$$

This uses nothing but exchangeability (Vovk et al., 2005). Positive or signed, the marginal number is the same. The sign enters only when we ask what the certificate hides.

4 The conditional-coverage decomposition

Condition on the realised width. Write $H(s, q) = \mathbb{P}(S_{n+1} \leq s \mid \hat{q} = q)$, so coverage conditional on the width is the diagonal $c(q) = H(q, q)$.

Lemma 1 (Conditional-coverage slope)**.** *If $(S_{n+1}, \hat{q})$ has a joint density, then*

$$c'(q) = \underbrace{f_{S_{n+1}\mid\hat{q}}(q \mid q)}_{\text{(i) threshold}} + \underbrace{\partial_q\, \mathbb{P}(S_{n+1} \leq q \mid \hat{q} = q)}_{\text{(ii) dependence}}. \tag{2}$$

Term (i) *is non-negative. Term* (ii) *is ≤ 0 when S_{n+1} is stochastically increasing in $\hat{q}$, $= 0$ under independence, and ≥ 0 when stochastically decreasing.*

Proof. Total derivative of $H(q,q)$ along the diagonal: $c'(q) = \partial_s H(s,q)\big|_{s=q} + \partial_q H(s,q)\big|_{s=q}$. The first equals $f_{S_{n+1}|\hat{q}}(q \mid q) \geq 0$. For the second, $\partial_q H(s,q) \leq 0$ for all s exactly when $\mathbb{P}(S_{n+1} \leq s \mid \hat{q} = q)$ is non-increasing in q, the definition of S_{n+1} being stochastically increasing in $\hat{q}$. The reverse inequality is stochastic decrease. □

Remark 1 (The independence case is classical)**.** *Under independence term (ii) vanishes and $c(q) = F(q)$, the test-score CDF, which already rises with the width purely from estimation. This is the known law of calibration-conditional coverage: $F(\hat{q}) = F(S_{(k)}) \sim \mathrm{Beta}(k, n-k+1)$ (Vovk, 2012; Marques F., 2024; Angelopoulos and Bates, 2023). Lemma 1 is its extension to dependent scores, with term (ii) the new piece.*

Remark 2 (Reading the sign)**.** *The width $\hat{q} = S_{(k)}$ is a coordinatewise non-decreasing function of the calibration scores, and the test score is disjoint from them. A positive (proper) de Finetti mixture makes the score vector* associated *(Esary et al., 1967); association is preserved by monotone maps, so $(\hat{q}, S_{n+1})$ are positively dependent, term (ii) is ≤ 0, and it cancels against the threshold term—conformal is* conditionally adaptive*, flatter than independence, because pooling the calibration residuals silently conditions on the realised world. A signed, negatively-correlated law makes the vector* negatively associated *(Joag-Dev and Proschan, 1983)—the property of without-replacement samples, multinomials, permutation and rank laws, and Dirichlet/compositional vectors, preserved under monotone maps of disjoint subsets—so term (ii) is ≥ 0 and adds to the threshold term: conformal is* conditionally anti-adaptive*. The calibration set anti-predicts the test, so coverage over-covers when the calibration scores were large and under-covers when small, even as the marginal stays pinned. The discrete (order-statistic) version replaces the derivative by a difference and behaves identically. The density assumption is for readability.*

The three regimes are confirmed numerically by `check_sign.py`, which runs split conformal in an independent, a positively-dependent, and a without-replacement world: the marginal coverage holds at $1-\alpha$ in all three, while coverage binned by the calibration sample stays flat under positive dependence and fans steeply under negative.

The reading is a go/no-go on conformal's coverage. With independent or positively-dependent data the certificate is a fair guide point by point, and the interval even tightens toward the case at hand. With negatively-dependent data the same number holds on average while the true coverage swings high and low, so it is close to useless per case. What carries information there is the latent you would estimate, not the conformal wrapper. Check the sign first. Positive, trust the coverage. Negative, model instead.

5 Why the signed corner is not exotic: constrained scores

The negative corner arises whenever the nonconformity score is defined *relative to the batch*, because a linear constraint forces it. De-meaning ($\sum_i s_i = 0$), differencing, ranks or percentiles-within-batch, and compositional or budgeted targets all impose such a constraint, and the constraint *is* the negative association (Joag-Dev and Proschan, 1983). For these scores conformal remains marginally valid—if the relativisation is symmetric across calibration and test, exchangeability is preserved—but its conditional coverage is the fanning, anti-adaptive kind of Lemma 1.

A caution: if the relativisation treats the test point asymmetrically (de-meaning by a training-only mean, say), exchangeability itself fails and even the marginal guarantee is no longer assured. This is the familiar reason split conformal uses out-of-fold rather than in-sample residuals (Lei et al., 2018); in-sample regression residuals are both constrained ($\sum_i e_i = 0$) and heteroscedastic (variance $1 - h_{ii}$), and so are not exchangeable to begin with.

Remark 3 (Rank, choice, and market-implied scores)**.** *The signed corner is the native habitat of contest data. In a Thurstone or Luce choice model a field of competitors carries latent values and the observed outcome is a winner or a full ranking. The winner indicator is a single multinomial trial, and the rank vector a permutation law, both* negatively associated *(Joag-Dev and Proschan, 1983). Any nonconformity score read off such a field—a rank, a within-field relative score, or a market-implied win probability—therefore sits in the negatively-associated regime above, where the marginal certificate is least informative about conditional reliability. The converse direction is the modelling the certificate cannot supply: recovering latent means from observed prices or winning probabilities (Cotton, 2021) is exactly the de Finetti latent-recovery of Section 2. In betting terms the market price is the crowd's Q, the recovered latent mean is an estimate of the true P, and the divergence between them is the rent of the companion paper. So the same contest data that lands conformal in its anti-adaptive corner is also where the de Finetti latent, once estimated, pays.*

6 Related work

The marginal-coverage fact is standard (Vovk et al., 2005). The Beta law for coverage given the calibration set is Vovk (2012), refined by Marques F. (2024); training-conditional coverage is Bian and Barber (2023). Magnitude-based bounds for dependent data are Barber et al. (2023) and Oliveira et al. (2024). Association and negative association are Esary et al. (1967) and Joag-Dev and Proschan (1983); finite and signed de Finetti are Diaconis and Freedman (1980) and Kerns and Székely (2006); Leonetti (2018).

Barber and Pananjady (2026) are closest: they show conformal can over- or under-cover under temporal dependence. That account is marginal, it grades a departure by the size of the dependence rather than its sign, and it does not condition on the calibration set. The sign, and its identification with the sign of the finite de Finetti measure, is what this note adds.

de Finetti describes a law. Conformal runs without one. Using a de Finetti measure means estimating it, and that estimation is ill-posed: the latent is recovered from finite data, so in general one regularizes, shrinking toward the field. Conformal skips this, and the residual-information gap is the price of skipping it. A signed measure gives nothing to estimate from at all, so the representation tells you how conformal behaves rather than standing in for it. Datta et al. (2025) argue conformal is not a de Finetti conditional at all.

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